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Research article

Dynamic linkages and spillover effects of biodiversity risk in socially responsible investment and commodity markets

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ABSTRACT

This study employs a novel biodiversity risk measure, developed through textual analysis, to examine how biodiversity risk affects socially responsible investment (SRI) and commodity markets. Biodiversity-related financial risks, arising from ecosystem degradation, represent an emerging and underexplored dimension of market risk, particularly for investors seeking sustainability-aligned portfolios. Our analysis reveals that both SRI equity and commodity indices consistently exhibit negative time-varying correlations with biodiversity risk, with correlations as low as −0.62 for the FTSE4Good US 100 and -0.53 for the FTSE4Good Global 100. Similarly, commodities like silver, gold, crude oil, and wheat also show negative correlations with biodiversity risk. These findings indicate that neither asset class serves as a reliable hedge against biodiversity-related shocks. Furthermore, biodiversity risk has a significant long-term spillover effect on SRI equity and commodity market returns. As biodiversity risk increases, it strengthens the connectedness between these markets, thereby amplifying the transmission of risk across them. These findings highlight the need for new risk management strategies and regulatory frameworks that account for biodiversity risk, opening new research pathways in finance and environmental sustainability.

1. Introduction

The recent extremes of climate change events serve as a reminder that the global economy has not yet fully recovered from the pandemic and now faces the dual challenge of addressing climate and biodiversity loss. Whereas central banks across the globe have already acknowledged the biodiversity loss as a financial risk for the assets held by the financial institutions. What role climate finance plays in such a scenario remains a question of utmost interest recently. For instance, Cepni et al. (2022) hedge the climate risk with green assets, and Karolyi and Tobin-de la Puente (2023) emphasize the importance of financing nature, especially biodiversity. This paper breaks the ice by considering a comprehensive news-based measure of biodiversity risk developed by Giglio et al. (2023) on socially responsible investment (SRI) indices and commodities (silver, gold, crude oil, and wheat) to enrich the literature. We looked at whether SRI equity indices and commodity market indices can provide protection against biodiversity risk. The findings of our study indicate that neither of these indices can be considered safe havens against biodiversity risk because they consistently showed negative correlations with the biodiversity risk measure. Additionally, the study finds that SRI equity markets and commodity markets are highly interconnected, which means that shocks or changes in one market can affect others. This suggests that market movements can spread across different sectors. Moreover, results showed that when there were more concerns related biodiversity loss, it led to even stronger connections among SRI equity and commodity markets. This increased interconnectedness is likely due to greater uncertainty and tension in the markets caused by rising biodiversity risk.

Given the considerable economic risks arising from climate change-induced biodiversity loss, there is a burgeoning field of research focused on exploring its potential implications (Giglio et al., 2023; Starks, 2023). Moreover, global initiatives such as the ‘Convention on Biological Diversity’ and the ‘Kunming Declaration’, as highlighted during the 2022 UN Biodiversity Conference, have vigorously called for decisive action to safeguard biodiversity. These actions may usher in regulatory changes aimed at protecting biodiversity in the future, impacting financial market returns and investment decisions in sectors heavily reliant on natural resources, such as construction, agriculture, and the food and

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beverage industry (Ali et al., 2024; Carvalho et al., 2023). It is important to note that our study's findings corroborate recent developments in the literature, revealing a prevalent oversight in properly factoring biodiversity risk into asset pricing, with investors in high-biodiversity-risk industries often failing to consistently demand appropriate compensation (Garel et al., 2024; Kalhoro and Kyaw, 2024; Ma et al., 2024).

The intensifying impacts of climate change and biodiversity loss are becoming undeniable threats to both the global economy and the financial markets. As the planet witnesses increasingly frequent and severe environmental disruptions, the economic fallout from biodiversity loss is taking on a new urgency (Dasgupta and Treasury, 2022; IPBES, 2019). The degradation of ecosystems not only affects natural habitats but also undermines the foundation of industries that rely on these ecosystems, from agriculture and forestry to fisheries and tourism (Adekola and Mitchell, 2011; Kattel, 2022). Consequently, the risk that biodiversity loss poses to financial assets, institutions, and global market stability has gained attention. Central banks and financial regulators are beginning to acknowledge biodiversity as a material financial risk. For instance, Giglio et al. (2023) and Hudson (2024) emphasize that financial institutions, especially those with large investments in natural resource dependent sectors, are vulnerable to such emerging environmental risks.

Over the past, financial markets have responded to environmental risks by turning to climate finance mechanisms, but biodiversity loss introduces an entirely new dimension of risk that is distinct yet deeply engraved within the climate change. Biodiversity-related financial risks (BRFR) are multidimensional, encompassing both physical risks (such as the immediate destruction of ecosystems) and transition risks (which stem from shifting regulatory frameworks and market preferences towards sustainability) (Bosch, 2022; Kedward et al., 2023). The undervaluation of these risks poses a serious concern, potentially leading to mispriced assets and unexpected financial losses across sectors heavily reliant on biodiversity (OECD).[1]

Despite increasing recognition of BRFR, there remains a notable gap in understanding how these risks propagate through financial markets, particularly in sectors like socially responsible investments and commodity markets (Gardes-Landolfini et al., 2024). As these markets are often viewed as potential safe havens during periods of environmental volatility, understanding their relationship with biodiversity risk is increasingly important. For instance, Cepni et al. (2022) conclude that green assets can provide protection against climate risks. However, the role of SRI and commodity markets in safeguarding investments from biodiversity loss has not been thoroughly explored. The literature exhibits a concerning void regarding the fact that biodiversity loss, unlike other environmental threats, can result in irreversible harm, profoundly transforming entire ecosystems and economic frameworks (Cardinale et al., 2012; Dasgupta and Treasury, 2022; Kennedy et al., 2023; Monastersky, 2014).

This paper aims to address this gap by applying a comprehensive, news-based biodiversity risk measure developed by Giglio et al. (2023) to evaluate the relationship between biodiversity risk and key financial indices, specifically SRI equity indices and commodity markets, including silver, gold, crude oil, and wheat. SRI indices focus on environmental, social, and governance factors, which makes them forward-looking in addressing sustainability. Commodity markets are directly linked to natural resources, so they are sensitive to biodiversity loss.

The central argument of this study is that neither SRI equity indices nor commodity market indices provide reliable protection against biodiversity risk, even though they are often seen as safer investments (Chatziantoniou et al., 2022; Cunha et al., 2020). Our findings reveal that both types of indices exhibit consistently negative correlations with biodiversity risk, indicating that they are vulnerable to the same shocks affecting ecosystems and natural resources. Furthermore, we discover a high degree of interconnectedness between SRI and commodity markets, meaning that shocks in one market can easily propagate to the rest. This interconnectedness suggests that the risks associated with biodiversity loss are not contained within individual market sectors but have the potential to trigger widespread disruptions across the financial system. In particular, the study shows that negative biodiversity risk sentiments—driven by increasing awareness of environmental degradation—intensify market correlations, exacerbating the transmission of shocks between SRI and commodity markets. This finding aligns with the broader literature on market sentiment and environmental risk, as negative environmental news tends to increase market uncertainty and trigger more pronounced responses from investors (Capelle-Blancard and Petit, 2019; Ma et al., 2023; Schoenmaker, Schramade and Investment, 2019a). However, our research goes one step further by showing that this sentiment-driven volatility is particularly noticeable when biodiversity risk is included, indicating that financial markets are still unprepared to adequately manage these risks.

The contribution of this paper lies not only in its application of a novel biodiversity risk measure but also in its broader implications for financial market stability and sustainability. As biodiversity loss accelerates, financial markets will need to develop more sophisticated tools for assessing and managing these risks. This paper provides a foundation for future research into the complex interactions between biodiversity and financial markets, while also offering practical insights for policymakers and investors. By demonstrating the limitations of existing financial instruments in providing protection against biodiversity risk, we call attention to the need for more robust, biodiversity-sensitive financial products and regulatory frameworks.

In nutshell, we examine how socially responsible investments and commodity markets respond to biodiversity-related concerns by analyzing their time-varying correlations and interconnectedness when faced with biodiversity risk. The findings highlight the importance of integrating biodiversity considerations into financial models and investment strategies, as well as the urgent need for comprehensive policies that promote the preservation of biodiversity. The financial sector, like other industries, is inextricably linked to the health of the Earth's ecosystems, and failing to account for this relationship could lead to profound economic consequences. Our research contributes to the growing body of literature that seeks to reconcile financial market stability with environmental sustainability, emphasizing biodiversity risk as a critical factor that cannot be ignored. Thus, the policy implications of the study aid to exigency of integrating biodiversity risk into financial regulations and investment frameworks, role of incentives in promoting biodiversity-aligned financial instruments (i.e., green bonds and ESG portfolios), cross-sectoral collaboration for biodiversity financing and mitigating the systematic financial risks.

The remaining paper is outlined as follows: Section 2 provides brief review of literature; Section 3 presents data and methodology; Section 4 shows empirical results; Section 5 provides discussion and policy implications; Section 6 presents conclusion.

2. Literature review

After climate change, biodiversity loss has become a major risk for financial markets. However, biodiversity risk has not received as much attention in existing literature as climate change risks. Biodiversity risk can affect financial stability in two main ways, such as physical risk and transition risk. Physical risk arises from the destruction of ecosystems, while transition risk impacts financial markets through regulatory changes and shifts in consumer and investor preferences toward sustainability (Bosch, 2022; Kedward et al., 2023). Recently, Giglio et al. (2023) has developed a comprehensive news-based measure of biodiversity risk and determined how it evolved over time, which helps to quantify biodiversity risk effects on financial assets. Although

[1] Biodiversity-related risks to the financial sector, https://www.oecd.org/en/about/projects/biodiversity-related-risks-to-the-financial-sector.html.

biodiversity and finance have been of great interest in financial research, more research is required to understand how biodiversity risks affect investment and stability in financial markets.

SRI and commodities such as gold and oil are often believed to be safe investment options, especially during times of environmental stress. The existing studies on performance of green assets and commodity futures show that these assets can provide shield against certain environmental risks (Cepni et al., 2022; Hoque et al., 2024; Yao and Zhang, 2024). For instance, Jabeur et al. (2021) examined the relationship between worldwide environmental indexes, green energy resources, and stock markets. They found that lower crude oil prices are associated with higher values of global environmental indexes and green energy resources. Another study by El Ouadghiri, Guesmi, Peillex, and Ziegler (2021) showed that between 2004 and 2018, public awareness of environmental issues significantly positively impacted the returns of US sustainability indexes while negatively affecting conventional stock indexes. Over the past two decades, responsible investing has expanded rapidly, creating a sizable sector that includes alternative asset classes (Hoque et al., 2024; Shahid et al., 2023). This growth has led to the rise of companies focused on renewable energy and clean technologies (Shahid et al., 2023). Such companies are essential for meeting national goals and international commitments, like the Sustainable Development Goals (SDGs) and the Paris Agreement (Sachs et al., 2020).

Moreover, due to the close interdependence of financial markets, disruptions in one industry may have an impact on others. Biodiversity risk may strengthen these ties, increasing market volatility and spreading risks throughout industries (Capelle-Blancard and Petit, 2019). According Ma et al. (2023), negative environmental news frequently creates market uncertainty. Zeng et al. (2023) examined the return connectedness between clean energy indices and agricultural commodity market. They revealed that the degree of interconnection varies over time and is sensitive to significant shocks from macroeconomic news and market crises. Therefore, we argue that biodiversity risk may affect the interconnectedness among financial markets and contribute to market instability. Understanding these connections may assist investors and lawmakers in anticipating the broader implications of biodiversity loss. However, accords such as the Convention on Biological Diversity and the Kunming Declaration show that biodiversity loss is increasingly recognized as a financial risk (Carvalho et al., 2023). The threats to biodiversity are becoming more and more recognized by regulators and financial institutions, especially for sectors like forestry and agriculture that rely on natural resources (Ali et al., 2024). According to Gardes-Landolfini et al. (2024), regulatory frameworks that promote sustainable investments and improve risk management may help in the management of biodiversity risks.

Given the rising interest among researchers and policymakers in biodiversity loss and financial risks, this study examines how biodiversity risk affects socially responsible investments (SRI) and commodities, assessing whether these assets can shield investors from biodiversity-related risks. Additionally, we analyze the spillover impact of biodiversity risk on SRI equity and commodity market returns. We further explore the interconnectedness within and between SRI and commodity markets to determine if biodiversity risk intensifies these correlations, thereby amplifying risk transmission across markets.

3. Data and methodology

3.1. Data

This study investigates the impact of biodiversity risk on socially responsible investments (SRI) and commodity markets. It focuses on four SRI equity indices—FTSE4Good Europe 50, FTSE4Good Global 100, FTSE4Good UK 50, and FTSE4Good US 100—and four major commodities: Silver (XAG), Gold (XAU), Crude Oil (WTC), and Wheat (WHT). We use daily data spanning from December 2, 2014, to December 16, 2022. These indices were selected as they include companies from the US, UK, Europe, and global that meet specific Environmental, Social, and Governance (ESG) standards. This makes them ideal for understanding how biodiversity risks impact investments aligned with sustainable practices. The selected commodities are key resources linked to natural ecosystems, allowing us to assess how biodiversity threats influence traditional resource markets. Moreover, we have chosen these market indices, including Silver (XAG), Gold (XAU), Crude Oil (WTC), and Wheat (WHT based on their direct or indirect relationship with biodiversity risks and their importance in the global market. Scholars and market players have shown considerable interest in the financialization of commodity markets. These markets provide portfolio managers and international investors with new opportunities to diversify their holdings (Mensi et al., 2015).

It is noteworthy that commodities often attract investors during periods of stress in traditional financial markets. This is because economic shocks tend to affect stock and commodity prices in opposite ways (Silvennoinen and Thorp, 2013). Moreover, commodity markets share a close relationship with biodiversity risk. For instance, crude oil production contributes to deforestation and habitat destruction, resulting in significant environmental degradation (Butt et al., 2013; Hamilton, 2009). This underscores its importance in understanding the broader impacts of biodiversity loss. Similarly, silver and gold have long been viewed as safe-haven assets (Reboredo, 2013). However, their mining activities are associated with environmental concerns, including local pollution and soil degradation (Islam et al., 2022). Finally, Wheat, as an agricultural commodity, is deeply connected to biodiversity. Agriculture relies heavily on healthy ecosystems for key factors like soil fertility, crop resilience, and pest control (Perfecto et al., 2019). Biodiversity loss directly impacts these factors, making the agriculture sector highly vulnerable. For instance, changes in ecosystems can reduce the quality of soil or lead to increased pest outbreaks, both of which threaten food security (Altieri et al., 2015). To conclude, we selected these four commodities because they emphasize different dimensions of the relationship between economic activities and biodiversity loss. Wheat illustrates the direct effects of biodiversity loss on agriculture, silver and gold represent the dual roles of commodities as financial assets and environmental disruptors, and crude oil exemplifies the energy sector. Understanding these relationships is crucial for assessing the broader consequences of biodiversity risks in international markets.

Data sources include Refinitiv Eikon for SRI and commodity indices, and biodiversity risk data from www.biodiversityrisk.org/download/, based on Giglio et al. (2023) daily biodiversity risk index derived from New York Times news articles.[2] To develop this index, dictionary of biodiversity-related terms is created, including terms such as biodiversity, ecosystems, ecology, species, habitats, forests, deforestation, fauna, wetland, wildlife, coral, among others (Giglio et al., 2023). News articles were selected if they contained at least two biodiversity-related sentences using terms identified in the biodiversity dictionary. The news related to biodiversity can be positive or negative. In order to separate such news stories, Giglio et al. (2023) used a BERT model to determine if each selected sentence was positive, negative, or neutral. Sentences with positive sentiment were assigned a value of $+1$, negative sentences a value of -1, and neutral sentences a value of 0. Positive values indicate increased biodiversity risk, negative values suggest reduced risk, potentially due to positive developments, and zero represents a neutral level, with neither high nor low biodiversity risk.

3.2. Methodology

In this study, we apply Dynamic Conditional Correlation-Generalized Autoregressive Conditional Heteroskedasticity (DCC–GARCH) model developed by Engle (2002) to examine relationship between biodiversity risk, socially responsible investments and commodity markets. We

[2] See Giglio et al. (2023), p.9 & 10.

choose this model as it can track how correlations and volatilities changes over time, which makes it ideal for exploring dynamic relations. The DCC-GARCH model builds on the original GARCH model framework, which not only model volatility of individual variables but also model changing correlations between them (Kim et al., 2016). According to Sadorsky (2012), the DCC-GARCH model is empirically superior to other multivariate GARCH models. Moreover, this approach gives clearer picture of how relations evolve over time rather than assuming they remain constant.

The DCC-GARCH model calculates a variance-covariance matrix, H_t. This matrix measures the risk relationships between variables. The H_t is stated as follows:

$$H_t = D_t R_t D_t \tag{1}$$

where H_t represents the conditional variance matrix, showing how risk varies and interact with time t. D_t represents a diagonal matrix that contains standard deviation of each variable. Whereas R_t represents time-varying correlations between variables at time t. Moreover, the correlation matrix, R_t, is calculated as:

$$R_t = Q_t^{*-1} Q_t Q_t^{*-1} \tag{2}$$

where Q_t is a matrix that tracks dynamic correlations. Q_t^{*-1} is a diagonal matrix with the inverse square root of diagonal elements of Q_t. This stage ensures that R_t has 1s on its diagonal and all other values between -1 and 1. The Q_t matrix progresses over time as per following formula:

$$Q_t = (1 - \varnothing_1 - \varnothing_2)\overline{R} + \varnothing_1 \mu_{t-1}\mu_{t-1}^{\prime} + \varnothing_2 Q_{t-1} \tag{3}$$

where $\overline{R}$ is average correlation matrix, $\varnothing_1$ and $\varnothing_2$ are parameters that determine how much weight is given recent data versus past trends, and μ_t are standardized residuals from earlier calculations. For this study, the DCC-GARCH model is relevant as it captures both changing volatilities and correlations. These features are critical for understanding how biodiversity risk affects SRI equity and commodity returns over time. By providing time-varying correlations, the model highlights how these relationships shift, offering valuable insights for risk management.

For return spillovers between SRI equity and commodity markets, we use Diebold and Yilmaz (2012) model. This model measures spillover using an extended vector autoregression (VAR) framework rather than the Cholesky factor orthogonalization.[3] The model starts with an N-variable $VAR(p)$ model as:

$$X_t = \sum_{i=1}^{p} \Phi_i X_{t-1} + \varepsilon_t \tag{4}$$

Where $\varepsilon \sim (0, \Sigma)$ is a vector of (i.i.d.) residuals.

$X_t = \sum_{i=0}^{\infty} A_i \varepsilon_{t-1}$ is a moving average with A_i. An $N \times N$ matrix A_i is defined recursively as:

$A_i = \Phi_1 A_{i-1} + \Phi_2 A_{i-2} + \ldots + \Phi_2 A_{i-p}$ with $A_{i-p} = 0$ for $i < 0$.

The model breaks down the variance of forecast errors to show how much one market affects another. The variance decomposition formula is:

$$\theta_{ij}^{g}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\left(e_i^{\prime} A_h \sum e_j \right) \right)^2}{\sum_{h=0}^{H-1} \left(\left(e_i^{\prime} A_h \sum A_h^{\prime} e_i \right) \right)^2} \tag{5}$$

Where σ_{jj} is the standard deviation of error terms and e_i is a selection vector, with 1 as the ith element and 0 otherwise. $\sum_{j=1}^{N} \theta_{ij}^{g}(H) \neq 1$. Each element is normalized by the row sum.

$$\acute{\theta}_{ij}^{g}(H) = \frac{\theta_{ij}^{g}(H)}{\sum_{j=0}^{N} \theta_{ij}^{g}(H)} \tag{6}$$

[3] See, Diebold and Yilmaz (2009).

With $\sum_{j=0}^{N} \theta_{ij}^{g}(H) = 1$ and $\sum_{j=0}^{N} \acute{\theta}_{ij}^{g}(H) = N$.

The total spillover index measures how much uncertainty spreads across markets. It is calculated as:

$$S^{g}(H) = \frac{\sum_{\substack{i,j=1 \\ i \neq j}}^{N} \acute{\theta}_{ij}^{g}(H)}{\sum_{i,j=1}^{N} \acute{\theta}_{ij}^{g}(H)} \bullet 100 = \frac{\sum_{\substack{i,j=1 \\ i \neq j}}^{N} \acute{\theta}_{ij}^{g}(H)}{N} \bullet 100 \tag{7}$$

Directional spillover to market i from all other markets j is measured as:

$$S_{i\rightarrow}^{g}(H) = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{N} \acute{\theta}_{ij}^{g}(H)}{\sum_{i,j=1}^{N} \acute{\theta}_{ij}^{g}(H)} \bullet 100 = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{N} \acute{\theta}_{ij}^{g}(H)}{N} \bullet 100 \tag{8}$$

Directional spillover from market i to all other markets j is measured as:

$$S_{i\leftarrow}^{g}(H) = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{N} \acute{\theta}_{ji}^{g}(H)}{\sum_{i,j=1}^{N} \acute{\theta}_{ji}^{g}(H)} \bullet 100 = \frac{\sum_{\substack{j=1 \\ j \neq i}}^{N} \acute{\theta}_{ji}^{g}(H)}{N} \bullet 100 \tag{9}$$

Net spillover from market i to all other markets j is measured as:

$$S^{g} = S_{i\leftarrow}^{g} - S_{i\rightarrow}^{g} \tag{10}$$

4. Empirical results

In this section, we present the empirical findings from the dynamic conditional correlation GARCH (DCC-GARCH) model. We also look at how these markets are connected and what factors influence their relationships.

4.1. Part A: descriptive statistics, unconditional and dynamic conditional correlations, and volatility spillover analysis (DCC-GARCH)

Table 1 shows the descriptive statistics for our key variables. The mean value of biodiversity risk index is 0.554, indicating that biodiversity risk is usually higher during the study period. This high average indicates that biodiversity risk is a significant factor influencing the market within this timeframe. The daily mean returns for both SRI and commodity indices are close to zero, showing minimal average growth on a daily basis. Moreover, the skewness and kurtosis values reveal that the data does not follow a normal distribution, suggesting the presence of extreme values and asymmetric behavior in the returns. This deviation from normality justifies the choice of the GARCH model.

Table 2 demonstrates the correlation matrix, which shows that biodiversity risk generally has negative correlations with most SRI and commodity indices. The findings challenge the common view that these markets could serve as natural hedges against biodiversity risk. For instance, indices like the FTSE4Good Global 100 and FTSE4Good US 100, along with commodities such as silver (XAG), gold (XAU), and wheat (WHT), consistently show negative correlations with biodiversity risk. This suggests that when biodiversity concerns escalate, these assets do not provide cushion for investors. In sustainable finance literature, ESG-aligned assets are often seen as offering stability amid environmental risks (Chen et al., 2021; Liu et al., 2023). However, this result suggests that these assets may lack protective value specifically against biodiversity risk. The observed negative correlations may indicate a potential underpricing of biodiversity risk across financial markets, signaling that markets have not fully internalized the economic consequences of biodiversity decline (Giglio et al., 2023; Mace et al., 2012). The modern portfolio theory suggests that assets that negatively correlate with systemic risks are expected to diversify portfolios and reduce overall risk exposure (Markowitz, 1952). However, if biodiversity risk is not reflected in current asset pricing, it may create blind spots in

Table 1
Descriptive statistics.

Variable	Mean	SD	Max	Min	Median	Skewness	Kurtosis	Obs
Biodiversity risk	0.553	1.176	10.000	−4.000	0.0000	1.1370	8.537	2077
FTSE4Good Europe 50	−0.0001	0.012	0.072	−0.1945	0.0004	−3.0312	47.269	2077
FTSE4Good Global 100	0.00002	0.0164	0.1781	−0.5292	0.0004	−15.8284	530.880	2077
FTSE4Good UK 50	−0.000002	0.0106	0.0774	−0.1134	0.0001	−0.9533	15.8013	2077
FTSE4Good US 100	0.0001	0.0194	0.1989	−0.6530	0.0003	−18.1452	631.4756	2077
XAG	−0.0001	0.0188	0.0880	−0.3760	0.0003	−4.2953	85.2682	2077
XAU	0.00002	0.0133	0.1229	−0.4422	0.0004	−17.2624	587.7065	2077
WTC	0.0002	0.0383	0.8116	−0.3573	0.0004	4.3999	122.3714	2077
WHT	−0.0002	0.0481	1.0986	−0.8239	0.0000	3.8995	195.234	2077

Note: This table shows descriptive statistics. Biodiversity risk is biodiversity risk index. FTSE4Good Europe 50, FTSE4Good Global 100, FTSE4Good UK 50, and FTSE4Good US 100 are SRI indices. XAG (silver). XAU (gold), WTC (crude oil), and WHT (wheat) are commodity market indices. *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.

Table 2
Correlation matrix.

	Variable	1	2	3	4	5	6	7	8	9
1	Biodiversity	1								
2	FTSE4Good Europe 50	0.019	1							
3	FTSE4Good Global 100	−0.020	0.707	1						
4	FTSE4Good UK 50	0.030	0.873	0.539	1					
5	FTSE4Good US 100	−0.020	0.617	0.985	0.442	1				
6	XAG	−0.013	0.270	0.474	0.183	0.459	1			
7	XAU	−0.032	0.244	0.585	0.091	0.593	0.785	1		
8	WTC	0.025	0.225	0.281	0.196	0.267	0.249	0.202	1	
9	WHT	−0.006	0.144	0.272	0.038	0.283	0.188	0.289	0.016	1

Note: This table shows the unconditional correlation results between biodiversity risk and following financial market indices. Biodiversity variable refers to biodiversity risk index. FTSE4Good Europe 50, FTSE4Good Global 100, FTSE4Good UK 50, and FTSE4Good US 100 are SRI equity indices. XAG (silver). XAU (gold), WTC (crude oil), and WHT (wheat) are commodity markets indices.

portfolio risk management strategies, leaving investors vulnerable when biodiversity risks materialize. This highlights an area where current ESG frameworks might not align with actual risk, echoing concerns in recent literature that biodiversity-related financial risks are underappreciated or inadequately assessed (Deutz et al., 2020; IPBES, 2019; Kopnina et al., 2024).

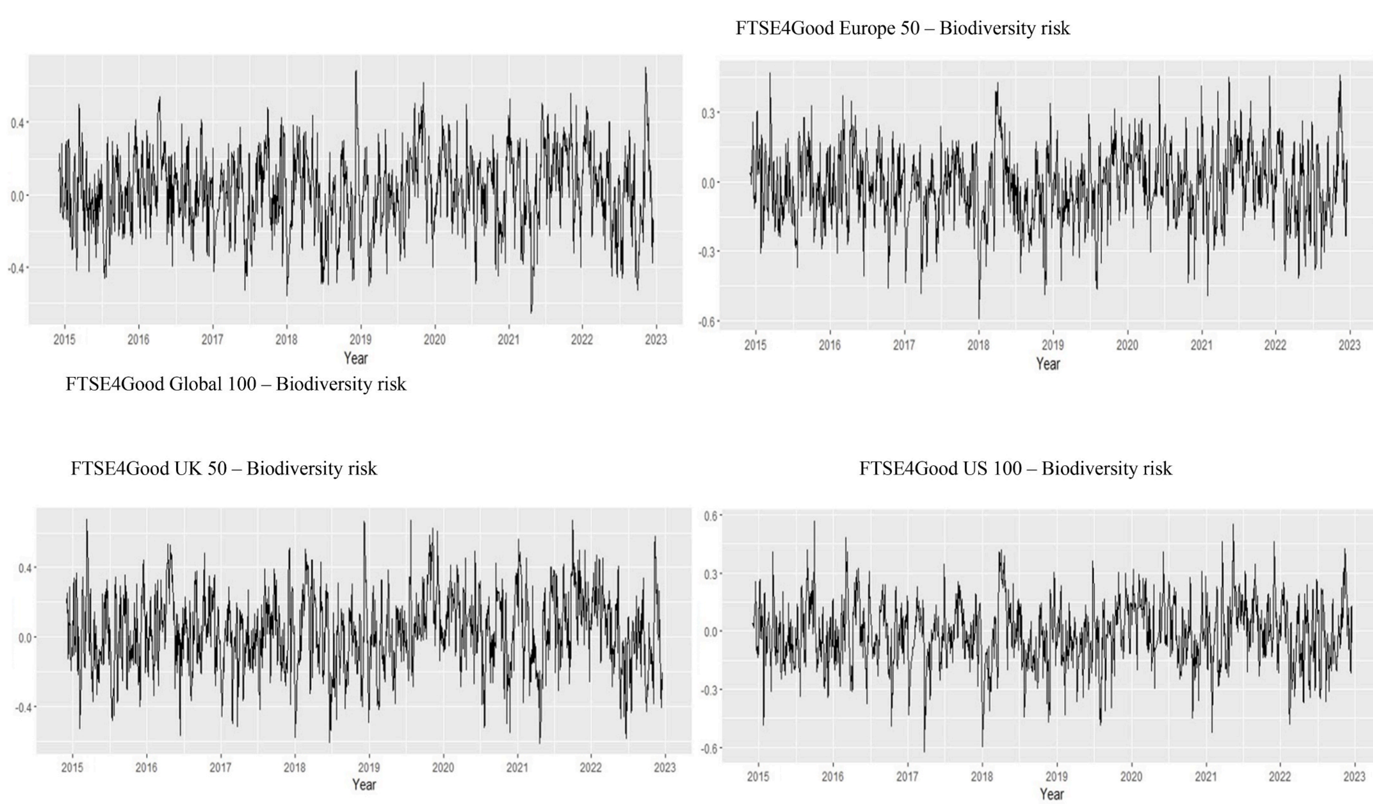

Fig. 1. Time-varying conditional correlation between log returns of four SRI equity indices and the biodiversity risk.

In contrast, a modest positive correlation observed in the FTSE4Good Europe 50, FTSE4Good UK 50, and crude oil indices, which suggests that certain regional markets may display varying sensitivity to biodiversity risk. For example, stricter environmental regulations in the European Union may increase the extent to which biodiversity factors influence asset prices, aligning with findings that regions with robust environmental policies tend to better incorporate biodiversity concerns into investment decisions (Hermoso et al., 2022). Moreover, this regional disparity underscores the importance of policy intervention and the role of regulatory alignment in standardizing biodiversity risk mitigation strategies.

The DCC-GARCH model's findings show that these relationships change over time, demonstrating how dynamically markets react to threats to biodiversity. Figs. 1 and 2 show that the negative correlations between biodiversity risk and indices like the FTSE4Good Global 100, FTSE4Good US 100, and important commodities like gold and crude oil tend to get stronger during times when biodiversity issues are more prominent, like significant environmental incidents or changes in regulatory policy. This implies that these markets may not provide investors with risk protection as biodiversity issues grow, casting doubt on the known stability of these assets in the face of biodiversity risk.

The fact that DCC-GARCH model permits correlations to adapt to changing situations, in contrast to classic models, emphasizes that biodiversity risk is not static but rather varies with changes in the environment and in policy. This variation is consistent with our previous results that showed negative correlations between biodiversity risk and these indices. This analysis also demonstrates that these relationships not only stay negative but also get stronger during periods of greater attention on biodiversity—possibly as a result of shifting investor attitude or enhanced market knowledge in reaction to news or legislation pertaining to biodiversity.

This changing correlation pattern shows that biodiversity risk impacts markets much like other types of systemic risks. This response aligns with portfolio theory, which stresses the need to include important risk factors in hedging and diversification strategies. The observed changes reveal possible gaps in current sustainability frameworks, especially concerning biodiversity risk. Our findings suggest that investors who rely on traditional hedging may not be fully protected against biodiversity risk, particularly during times of high biodiversity concern.

The summary statistics of time-varying correlations reported in Table 3 reveal that biodiversity risk has a negative average correlation with the FTSE4Good Global 100, FTSE4Good US 100, and all commodity indices, including silver (XAG), gold (XAU), crude oil (WTC), and wheat (WHT). This finding indicates that as concerns about biodiversity increase, these assets tend to perform poorly, suggesting they do not act as effective hedges against biodiversity risk.

Interestingly, the average correlation between biodiversity risk and the FTSE4Good Europe 50 index is positive, which is consistent with our previous results. This discrepancy suggests that European markets may respond differently to biodiversity issues, potentially due to stronger regulatory frameworks or greater investor awareness of environmental

Table 3
Time-varying correlations (DCC-GARCH).

	Mean	SD	Min	Max
FTSE4Good Europe 50/Biodiversity	0.023	0.221	−0.653	0.703
FTSE4Good Global 100/Biodiversity	−0.016	0.160	−0.591	0.470
FTSE4Good UK 50/Biodiversity	0.031	0.224	−0.614	0.677
FTSE4Good US 100/Biodiversity	−0.015	0.167	−0.622	0.567
XAG/Biodiversity	−0.025	0.039	−0.156	0.116
XAU/Biodiversity	−0.035	0.037	−0.157	0.113
WTC/Biodiversity	−0.006	0.043	−0.125	0.137
WHT/Biodiversity	−0.019	0.040	−0.162	0.079

Note: This table shows pairwise dynamic conditional correlation between SRI equity indices, commodity indices, and biodiversity risk index.

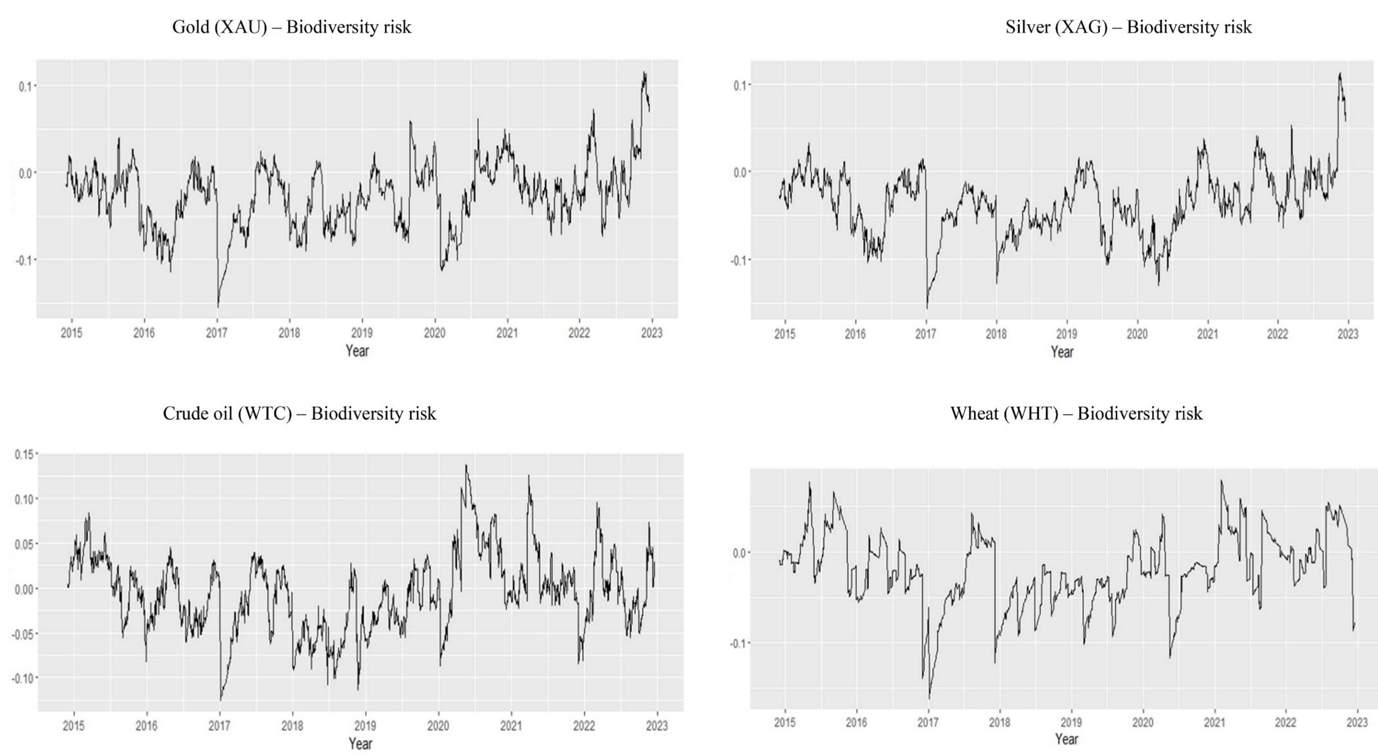

Fig. 2. Time-varying conditional correlation between the log return of commodity indices (Gold, Silver, Crude oil, and Wheat) and the biodiversity risk.

factors. The overall negative correlations with commodities imply that these assets may not provide safe havens for investors concerned about biodiversity loss. This suggests that traditional safe-haven assets, such as gold and oil, may not adequately protect against the unique risks posed by biodiversity decline.

Table 4 presents the results of the DCC-GARCH parameter estimates, which analyze the spillover effects of biodiversity risk on different asset classes. In Panel A, which focuses on Socially Responsible Investment (SRI) equity indices, the parameter θ_2 (DCC b1) is statistically significant at 5%. This finding suggests that biodiversity risk has a significant long-term spillover impact on SRI equity returns, meaning that as concerns regarding biodiversity increase, SRI equities are likely to experience corresponding changes in returns. However, θ_1 (DCC a1) is not statistically significant, indicating that short-term spillover effects from biodiversity risk to SRI equity indices are not supported by the data. This aligns with existing literature suggesting that while environmental concerns influence long-term investment strategies, their immediate effects may not be as pronounced (Fulton et al., 2012; Schoenmaker, Schramade and Investment, 2019b).

In Panel B of Table 4, the results for biodiversity spillover to commodity indices show that the parameter θ_2 (DCC b1) is statistically significant at 1%, suggesting that biodiversity risk also has a long-term spillover impact on commodity returns. The heightened significance at this level suggests a strong connection between biodiversity risk and the volatility of commodity markets, reinforcing previous findings that commodity prices are increasingly influenced by environmental and sustainability factors (Gallic and Vermandel, 2020; Li et al., 2022). The significance of θ_2 in both panels emphasizes the importance of biodiversity risk in shaping the returns and volatility of both SRI equity indices and commodity markets. Given that biodiversity loss can lead to increased market volatility and uncertainty, investors should consider these risks when constructing portfolios. Therefore, integrating biodiversity risk into financial analysis becomes essential for achieving sustainable investment outcomes and mitigating potential losses.

4.2. Part B: dynamic connectedness among markets and their determinants

Table 5 displays results of the interconnectedness among different SRI equity indices, using spillover percentages to highlight how shocks in one market affect others. For instance, on average, 42.17% of directional spillovers are transmitted across SRI markets, meaning a sizable portion of a shock in one market impacts others, reflecting their interconnected nature. Meanwhile, the remaining 57.83% of the shock effect stays within the originating market, suggesting that these SRI markets have some internal resilience, enabling them to absorb shocks to a certain degree without fully transmitting them externally.

Looking further into individual indices, FT4EU (FTSE4Good Europe 50) is primarily a net transmitter, meaning it often sends more spillovers than it receives, thereby influencing other indices. FT4G (FTSE4Good Global 100) and FT4US (FTSE4Good US 100), on the other hand, act as net receivers, indicating that they are generally more impacted by spillovers from other indices than they transmit themselves. FT4UK (FTSE4Good UK 50) shows a balanced role, acting as both a transmitter and receiver, although with a slightly positive net contribution to others.

The Total Connectedness Index (TCI) of 126.52% emphasizes the high level of integration among these SRI indices, signifying that changes in one index have a strong potential to influence others. Additionally, the Net Pairwise Directional Connectedness (NPDC) value of 42.17 across these indices further supports that they are intricately connected, with significant directional spillover activity occurring among them.

Table 6 extends this analysis to show that, on average, 35.03% of spillover effects travel between SRI and commodity markets, while 64.97% of the impact remains within the originating market. This means that although each market has some internal resilience, there is still a significant level of connection among them. When shocks happen in one market, they are likely to affect others.

The FTSE4Good Europe 50 index (FT4EU) is the strongest

Table 5
Dynamic connectedness of SRI equity returns indices.

	FT4EU	FT4G	FT4UK	FT4US	FROM others
FT4EU	93.03	1.24	4.65	1.08	6.97
FT4G	9.5	41.08	5.39	44.03	58.92
FT4UK	7.26	1.1	90.81	0.83	9.19
FT4US	7.56	40.15	3.73	48.56	51.44
TO others	24.32	42.48	13.77	45.95	126.52
NET	17.35	−16.44	4.59	−5.5	TCI
NPDC	0	3	1	2	**42.17**

Note: "TO others" represents how much influence each index has on the other indices. "FROM others" shows how much influence each index receives from all the other indices. NET represents the net connectedness (the difference between the total influence sent out and the total influence received). TCI stands for Total Connectedness Index. NPDC means Net Pairwise Directional Connectedness.

Table 4
DCC-GARCH model results.

Panel A: Biodiversity spillover to SRIs					
	Biodiversity	FTSE4Good Europe 50	FTSE4Good Global 100	FTSE4Good UK 50	FTSE4Good US 100
Ω	0.451***	0.000	0.001***	0.000	0.001
μ	0.036	0.000	0.000	0.000***	0.000
α	0.078	0.064	0.330	0.145***	0.298
β	0.898***	0.914	0.657***	0.750***	0.673
DCC Estimation					
θ_1	θ_2	Akaike	Bayes	Shibata	Hannan-Quinn
0.170	0.792**	−19.271	−19.185	−19.272	−19.239
Panel B: Biodiversity spillover to commodity market					
	Biodiversity	XAG	XAU	WTC	WHT
Ω	0.451***	0.000	0.000	0.000	0.001
μ	0.036	0.000***	0.000***	0.000**	0.000
α	0.078	0.050***	0.000	0.141***	0.058
β	0.898***	0.933***	0.968***	0.815***	0.941***
DCC Estimation					
θ_1	θ_2	Akaike	Bayes	Shibata	Hannan-Quinn
0.010	0.974***	−16.180	−16.093	−16.180	−16.148

Note: *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.

Table 6
Dynamic connectedness of SRI equity returns indices and commodity indices.

	FT4EU	FT4G	FT4UK	FT4US	XAG	XAU	WTC	WHT	FROM others
FT4EU	86.63	1.63	8.46	1.43	0.34	0.47	0.97	0.08	13.37
FT4G	9.77	29.8	5.22	32.4	8.15	8.12	4.89	1.65	70.2
FT4UK	12.97	1.34	83.08	1.02	0.28	0.11	1.19	0.02	16.92
FT4US	7.93	30.02	3.68	35.8	7.71	8.36	4.72	1.77	64.2
XAG	1.44	5.75	0.76	5.87	66.67	15.36	3.71	0.46	33.33
XAU	1.22	3.56	0.18	3.96	9.56	79.2	1.96	0.35	20.8
WTC	1.42	1.21	1.15	1.26	1.3	1.1	92.53	0.03	7.47
WHT	1.51	5.58	0.28	6.48	2.14	2.67	0.27	81.06	18.94
TO other	36.26	49.09	19.73	52.42	29.48	36.19	17.71	4.35	245.22
NET	22.89	−21.1	2.81	−11.78	−3.85	15.39	10.24	−14.59	TCI
NPDC	0	6	2	5	4	3	1	7	**35.03**

Note: "TO others" represents how much influence each index has on the other indices. "FROM others" shows how much influence each index receives from all the other indices. NET represents the net connectedness (the difference between the total influence sent out and the total influence received). TCI stands for Total Connectedness Index. NPDC means Net Pairwise Directional Connectedness.

transmitter of these spillovers, with 22.89% net connectedness, suggesting it plays a key role in spreading risk to other markets. This could be due to Europe's strict environmental policies and high awareness of biodiversity issues, making European markets more responsive to biodiversity risks, which then ripple through to other sectors. On the other hand, Gold (XAU) and Crude Oil (WTC) also emerge as major transmitters, with net connectedness values of 15.39% and 10.24%, respectively. Commodities like gold and oil are sensitive to global economic changes and environmental factors, which makes them effective conduits for transmitting risk across markets.

These findings suggest that certain European SRI indices and key commodities are central in the network of risk transmission. This interconnected structure means that a biodiversity risk event in one of these primary markets could quickly influence the entire financial system, impacting both SRI and commodity sectors.

Next, we look at how biodiversity risk, based on the textual analysis by Giglio et al. (2023), affects interconnectedness among SRI equity indices and interconnectedness between SRI equity and commodity indices. To control for other factors, we include daily economic policy uncertainty (EPU) data from the UK and the US, along with market volatility indices (VIX, EVZ, MOVE), and the economic impacts of Covid-19. The VIX (CBOE Volatility Index) shows expected volatility for the S&P 500 index over the next 30 days. The EVZ (CBOE Eurocurrency Volatility Index) measures volatility for the EUR/USD exchange rate, while the MOVE (Merrill Lynch Option Volatility Estimate) reflects volatility in the bond market. We treat Covid-19 as a binary variable, assigning a value of 1 from December 2, 2019, to September 30, 2021, and a value of 0 at other times.

Table 7 displays how SRI equity and commodity markets are connected and how this relationship interacts with uncertainty factors, such as biodiversity risk, EPU UK, EPU US, VIX, EVZ, MOVE, and the COVID-19 pandemic. In Model 1, the results indicate that a biodiversity risk is positively associated with interconnectedness between SRI equity indices and commodity markets at a 10% significance level. This implies that rising biodiversity risks lead to tighter connections between these markets, likely due to heightened investor uncertainty and greater risk aversion. As biodiversity issues grow, investors may respond by adjusting their positions across both SRI equities and commodities, amplifying the co-movement between these markets and increasing systemic risk.

However, Model 2 indicates that biodiversity risk does not significantly impact interconnectedness within SRI equity markets alone. This result suggests that biodiversity concerns may be more influential in linking SRI equities with broader markets, like commodities, than in driving internal changes within the SRI equity market itself.

Table 7
Regression results of contributing factors to total connectedness of SRI equity and commodity markets.

	(Model 1)	(Model 2
	TCI – SRI equity & Commodities	TCI – SRI equity
BRI	0.0229*	−0.0018
	(1.76)	(-0.67)
EPU_UK	0.0004***	0.0000
	(2.75)	(1.59)
EPU_US	0.0001	−0.0002
	(0.29)	(-1.43)
VIX	−0.0080	0.0133***
	(-1.11)	(4.7)
EVZ	0.0352***	0.0108***
	(2.73)	(4.34)
MOVE	−0.0004	−0.0016***
	(-0.44)	(-4.39)
Covid19	−0.0736*	−0.0340***
	(-1.78)	(-3.16)
Constant	35.63***	41.98***
	(178.28)	(942.27)
N	2001	2001
R-sq	0.078	0.186

Note: This table shows the results of the regression estimates. In Model 1, the dependent variable is the TCI (Total Connectedness Index) of SRI equity and commodity indices, while the independent variables are the Biodiversity Risk Index (BRI), EPU_UK (Economic Policy Uncertainty Index for the UK), EPU_US (Economic Policy Uncertainty Index for the US), VIX (CBOE Volatility Index), EVZ (CBOE Eurocurrency Volatility Index), MOVE (Merrill Lynch Option Volatility Estimate), and Covid-19 (a binary variable). In Model 2, the dependent variable is the TCI of SRI equity indices, and the independent variables remain the same. The data frequency is daily. We include the month as time fixed effect in the regression to account for potential seasonal effects. The standard errors are robust to account for potential heteroscedasticity in the data. t-statistics are shown in parentheses below the estimated coefficients. *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels, respectively.

5. Discussion and policy implications

This study explores the time-varying relationship between biodiversity risk, socially responsible investment, and commodity markets. The results show that biodiversity risk has a negative correlation with most SRI and commodity indices. This pattern challenges the usual assumption that these assets protect against environmental risks. Assets like the FTSE4Good Global 100, FTSE4Good US 100, and commodities such as gold, silver, and wheat consistently show this negative relationship with biodiversity risk. This suggests that as biodiversity concerns grow, these assets do not provide a protective buffer, contrasting with prior views that see ESG-focused assets as more resilient to environmental risks. DCC-GARCH model further shows that the correlation between biodiversity risk and SRI equity and commodity market indices is negative, time-varying, and rises during periods of high biodiversity concern, such as environmental crises or regulatory changes. This pattern suggests that biodiversity risk affects the whole market over time and that neither SRI nor commodity assets offer effective protection against it. This study supports earlier research suggesting that biodiversity risk is often underpriced, meaning that markets may not fully consider the financial impacts of biodiversity loss (Giglio et al., 2023; Kedward et al., 2023). This study adds to sustainable finance research by questioning the ability of ESG frameworks to manage biodiversity risks. It shows the need for a combined approach to create consistent global regulations, especially in areas with limited standards on biodiversity risk. Consistent regulations could help form a standard approach for managing these risks. Additionally, the study finds that there are major spillover effects between SRI and commodity markets during times of high biodiversity concern. This highlights the need for policies to reduce systemic risks and keep financial markets stable.

Our study also provides some implications for both investors and policymakers. For investors, the findings show the importance of including biodiversity risk in ESG-aligned portfolios. Since SRI and commodity indices may not protect against biodiversity-related events, investors might need to reconsider their diversification strategies. Relying on SRI and commodity assets as hedges may expose portfolios to biodiversity-related risks. To manage these risks, investors should look for assets that respond well to biodiversity concerns or broaden diversification to stay stable during environmental challenges. For policymakers, the findings suggest the need for regulations that better include biodiversity risks in financial markets. Some European indices showed a slight positive correlation with biodiversity risk, possibly due to stricter environmental policies in that region. This suggests that regulatory actions in regions with fewer policies could help bring biodiversity risks to the surface. Policymakers should also encourage consistent disclosures and motivate companies to report their biodiversity impacts, helping investors make better decisions and protect their investments from biodiversity risks.

6. Conclusion

Biodiversity loss is an urgent issue that affects both the environment and the economy. Coordinated global actions, like the 2021 Kunming Declaration and the 2022 UN Biodiversity Conference, have focused on protecting biodiversity. However, these efforts may create new risks for financial investors, as shifts in environmental rules and sustainable practices could impact markets and investment choices. To address the rising importance of biodiversity risk, this research examines how SRI equity indices and commodity markets respond to biodiversity risk and whether these markets can provide investors protection against this risk. We find that neither SRI equity indices nor commodity market indices provide investors protection against biodiversity risk, as indicated by consistently negative time-varying correlations with the biodiversity risk index. Moreover, interconnectedness among SRI equity markets and between SRI and commodity markets was pronounced, suggesting that shocks and market movements could spread across sectors. In our regression analysis, we find that heightened biodiversity concerns led to stronger connections between SRI equities and commodity markets, likely because they increased uncertainty and tension.

In particular, the empirical results reinforce that biodiversity risk poses a significant and systemic threat to both SRI and commodity markets. Persistent negative correlations between biodiversity risk and key indices, such as −0.62 for the FTSE4Good US 100 and -0.53 for the FTSE4Good Global 100, suggest that these markets do not offer reliable hedges against biodiversity risk. Similarly, commodities like silver, gold, crude oil, and wheat exhibit negative correlations with biodiversity risk, indicating that they, too, offer limited protection during environmental uncertainty. Our dynamic connectedness analysis reveals that, on average, 42.17% of shocks originating in one SRI market are transmitted to others, while 35.03% of shocks propagate across both SRI and commodity markets, underscoring the potential for systemic risk. Furthermore, the DCC-GARCH model estimates suggest long-term spillover effects from biodiversity risk on both SRI equity and commodity returns, with the parameter θ_2 (DCC b_1) being statistically significant at the 5% and 1% levels, respectively.

These findings emphasize the need for policymakers and regulators to address the financial risks associated with biodiversity loss. The limited ability of SRI and commodity markets to hedge against biodiversity risk shows that this risk is not yet fully integrated into investment decision-making. Policymakers should establish clear guidelines and incentives to promote sustainability-focused investments, while encouraging the development of biodiversity-sensitive risk management tools. Additionally, new research should focus on biodiversity risk management across not only SRI equity markets but also other financial sectors, including commodity markets. Global coordination, innovative financial products, and enhanced regulatory frameworks would be essential to ensuring market stability and fostering long-term sustainability in the face of accelerating biodiversity loss.

7. Limitations and future directions

Although study uses the latest data and contemporary approach to reach the overall objectives of the study, but more granular data would have provided more precision generalized outcomes. Nonetheless, it opens up a future research direction for developing an advanced biodiversity risk metrics.

CRediT authorship contribution statement

Muhammad Ramzan Kalhoro: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Khalid Ahmed:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

Authors declare that there is not any conflict of interest.

Data availability

Data will be made available on request.

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